

Title: Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Problem Statement

The paper addresses the limitation of parametric language models that store knowledge only within fixed model parameters, making them difficult to update and prone to hallucinations. The study proposes a retrieval-based approach to improve factual accuracy and performance on knowledge-intensive NLP tasks by accessing external knowledge sources dynamically.

Algorithms Used

The study introduces the **Retrieval-Augmented Generation (RAG)** framework, which combines dense passage retrieval (DPR) with sequence-to-sequence transformer models such as BART. The model retrieves relevant documents from external knowledge sources and integrates them during response generation.

Datasets

The model is evaluated on knowledge-intensive tasks including open-domain question answering datasets such as Natural Questions, WebQuestions, CuratedTREC, and FEVER fact verification datasets. Wikipedia is used as the external knowledge source.

Model Training and Testing

The retriever and generator components are jointly trained. The retriever selects relevant passages using dense vector similarity, and the generator produces responses conditioned on retrieved documents. Performance is evaluated using answer accuracy and factual correctness.

Results

The RAG model significantly improves performance on knowledge-intensive tasks compared to parametric-only models. It produces more factual and interpretable outputs while allowing knowledge updates without retraining the entire model.

Conclusions

The authors conclude that combining retrieval with generation improves factual accuracy, interpretability, and scalability of language models.

Open Questions

Future work includes improving retrieval efficiency, handling noisy retrieved documents, and improving integration between retrieved knowledge and generation.

Relevance to Our Team

This paper directly supports our project's RAG pipeline for clinical document analysis. The retrieval-based framework aligns with our approach of retrieving relevant medical text chunks and generating evidence-based responses from external knowledge sources.

Title: REALM: Retrieval-Augmented Language Model Pre-Training

Problem Statement

The paper addresses the challenge of storing and accessing factual knowledge in language models. Traditional models require retraining to update knowledge and struggle with knowledge-intensive tasks.

Algorithms Used

The study introduces **REALM (Retrieval-Augmented Language Model)**, which integrates neural retrieval with masked language model pretraining. It jointly trains a retriever and language model to access external documents during training.

Datasets

The model uses large text corpora including Wikipedia as the knowledge base and evaluates performance on open-domain question answering benchmarks.

Model Training and Testing

The model retrieves relevant documents using dense embeddings and incorporates retrieved text into masked language modeling objectives. Training jointly optimizes retrieval and prediction performance.

Results

REALM significantly improves performance on knowledge-intensive tasks and reduces dependency on large parameter sizes by leveraging external knowledge.

Conclusions

The authors conclude that retrieval during pretraining enables language models to learn knowledge access mechanisms and improves factual reasoning.

Open Questions

Future work includes scaling retrieval databases and improving retrieval quality for complex reasoning tasks.

Relevance to Our Team

REALM provides foundational support for our system's embedding-based retrieval and knowledge access mechanisms. It justifies using external medical knowledge sources rather than storing all knowledge in model parameters.

Title: Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering

Problem Statement

The paper addresses limitations of generative models in open-domain question answering, where models struggle to produce accurate responses without access to external knowledge.

Algorithms Used

The study combines passage retrieval methods with generative transformer models. It uses dense retrieval methods to fetch relevant passages and conditions answer generation on retrieved text.

Datasets

The model is evaluated on open-domain question answering datasets including Natural Questions and TriviaQA.

Model Training and Testing

The system retrieves relevant passages using embedding similarity and feeds them into a generative model. Training optimizes answer generation conditioned on retrieved context.

Results

The approach improves answer accuracy and factual consistency compared to models without retrieval.

Conclusions

The authors conclude that integrating retrieval with generation improves knowledge access and enhances performance on question answering tasks.

Open Questions

Future work includes improving passage ranking and handling irrelevant retrieval results.

Relevance to Our Team

This paper supports our design of retrieving relevant medical document chunks before generating responses. It validates the importance of retrieval-based question answering for evidence-based systems.

Title: Atlas: Few-shot Learning with Retrieval Augmented Language Models

Problem Statement

The paper addresses the challenge of few-shot learning in language models, where models require large training data and struggle to generalize with limited examples.

Algorithms Used

The study introduces the **Atlas framework**, which integrates retrieval-based memory with language models to support few-shot learning. It combines dense retrieval with transformer-based generation.

Datasets

The model is evaluated on multiple NLP benchmarks including question answering, reasoning, and knowledge-intensive tasks using large-scale text corpora.

Model Training and Testing

The model retrieves relevant examples from external databases and conditions predictions on retrieved information. Training involves optimizing retrieval and generation jointly.

Results

Atlas achieves strong few-shot performance and improves knowledge access without requiring extensive training data.

Conclusions

The authors conclude that retrieval-based augmentation improves generalization and reduces dependence on large training datasets.

Open Questions

Future work includes improving retrieval quality and scaling retrieval systems for large knowledge bases.

Relevance to Our Team

Atlas supports our system design where limited labeled clinical data can be supplemented using retrieval from external medical documents. It helps improve evidence-based response generation with minimal training data.